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Which Energy Expenditure Equation is Most Accurate in a Field-Based Setting?

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WHICH ENERGY EXPENDITURE EQUATION IS MOST ACCURATE IN A FIELD-
BASED SETTING?

by

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A plan B research project submitted in partial fulfillment

of the requirements for the degree

of

MASTERS OF SCIENCE

in

Kinesiology

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Abstract

Several energy expenditure prediction equations for walking on varied terrain exist; however, these predictions are often theoretical and have not undergone rigorous field testing for accuracy. Field research on this topic is sparse. The aim of this study was to determine if any existing energy expenditure prediction equations are accurate in a field-based setting with load and gradient being taken into account (i.e., under conditions that are most applicable to day hiking). This was done by directly measuring relative VO_2 (ml/kg/min) of study participants and comparing these direct measurements to the values generated from the existing prediction equations. Fifteen subjects (33% women) completed the study. Measurements of VO_2 were obtained from a portable metabolic unit while hiking a trail of varied terrain. Measured data were compared to the predicted values from the equations using correlations, ANOVA, and equivalence testing. It was found that none of the equations showed equivalence during downhill grades. Mean differences over the entire uphill and downhill trail, and all trail sections, ranged from -9.67 mL/kg/min to 3.40 mL/kg/min. The Ludlow equation showed equivalence over one uphill trail section, Dill showed equivalence over two, and the Pandolf, de Müllenheim, and Looney equations over three trail sections. In conclusion, the Pandolf, de Müllenheim, and Looney prediction equations are the best options for field use when comparing equivalence to measured VO_2 , with the Pandolf equation being the best option when looking at statistically significant differences.

Introduction

The ability to predict energy expenditure is important, enabling militaries, humanitarian and wildland firefighting organizations, as well as individuals to carefully plan what food to bring on any type of wilderness excursion (Looney et al., 2019). It is also useful to those in exercise programs, so that they can accurately stay within desired energy deficits or surpluses depending on the goals of the program. While there are widely used and accurate methods of obtaining energy expenditure, including metabolic gas analysis systems for lab-based settings and mobile metabolic units for field-based settings (Crouter et al., 2019; DeBlois et al., 2021; Guidetti et al., 2018; Perez-Suarez et al., 2018), these are expensive, require trained personnel, and are impractical for the general public to obtain energy expenditure data. The development of energy prediction equations bypasses the limitations of conventional methods, allowing for anyone to accurately predict their energy expenditure for whatever purposes they need.

There are several energy prediction equations, all with certain limitations and advantages. While many have been proven to be accurate and reliable, these validation studies were done in lab-based settings (Clavero-Jimeno et al., 2024; Ludlow & Weyand, 2016; Stoedefalke & Hawkins, 2022; Xue et al., 2021); consequently, generalizability to outside settings is limited. In real-world settings, people carry an additional external load with them like a backpack containing water, extra clothing, etc., and most scenarios also take place with some sort of gradient. Previous lab-based studies did not have these characteristics. There is a lack of research validating these equations in a field-based setting generalizable to day hiking; only one study by Weyand et al. (2021) has done this, and it was criticized by others for questionable post-hoc manipulation of results (Looney et al., 2022*b*).

The aim of this study was to find which of five energy expenditure prediction equations (de Müllenheim et al., 2016; Dill, 1965; Looney et al., 2022a; Ludlow & Weyand, 2017; Pandolf et al., 1977; Weyand et al., 2021) are most accurate in a field-based setting during trail hiking with a light load and gradient being taken into account.

Methodology

Five equations widely used in the literature incorporate surface gradient into their estimations (de Müllenheim et al., 2016; Dill, 1965; Looney et al., 2022a; Ludlow & Weyand, 2017; Pandolf et al., 1977, Weyand et al., 2021); those ones were used to compare to measured results (Table 1 & Table 2). It should be noted that the Dill equation is hard to elucidate from the original paper (Dill, 1965), but is commonly used in the literature as represented by Weyand et al. (2021). In its original form, the Dill equation – commonly referred to as the ACSM equation – only allows estimation when grade is either zero or positive (Dill, 1965; Weyand et al., 2021). When grade is negative, this value either needs to be treated as zero, or a 0.73 adjustment can be included in the moving portion of the equation while keeping grade 0 like done by Weyand et al. (2021). The equation from de Müllenheim et al. (2016) also has this same problem, and the same adjustment can be made for it as well. A simple linear adjustment like this likely underestimates energy expenditure needs at steeper negative grades (Looney et al., 2019; Santee et al., 2003), but as the adjustment has been used elsewhere (Weyand et al. 2021) and would likely be more accurate than treating a negative grade as zero, it was included here. Potter et al. (2018) argued that simply treating negative grade as 0 is more accurate on shallower downhills and applying the 0.73 adjustment to these prediction equations would be the most accurate on steeper downhills, but we used the 0.73 adjustment here to simplify the Dill and de Müllenheim equations.

The Looney equation, Pandolf equation, and Ludlow equation – commonly referred to as the Minimum Mechanics model – don't have these same limitations with negative gradient (Looney et al., 2019; Ludlow & Weyand, 2017; Pandolf et al., 1977), though the Pandolf equation is only able to accurately incorporate negative grades due to the correction factor added by Santee et al. (2003). Santee et al. (2003) developed this correction factor to fix the underestimation of energy expenditure the Pandolf equation originally had during downhill grades; this is implemented by subtracting the correction factor from the original Pandolf equation (see Table 1 & 2). It should be noted that Weyand et al. (2021) did not correctly use this correction factor in their field validation study, and both Looney et al. (2019) and Potter et al. (2018) also listed it incorrectly in their papers. Looney et al. (2022a) also never directly states what units their outcome measure is in. It was assumed to be in Watts/kg based on previous work (Looney et al., 2019).

	Grades	Equation
Ludlow	≥ 0	$VO_2 = RMR + [(m + l)/m]t[0.32g + 3.28 + (1 + 0.19g)2.66s^2]$
	< 0	$VO_2 = RMR + 0.73[(m + l)/m]t[0.32g + 3.28 + 2.66s^2]$
Looney		$Watts/kg = M_{rest} + [0.19 + t(1.78s^{0.58} + 0.27s^4 + (0.34sg(1 - 1.05^{1 - 1.1^{(g + 32)})))](1 + 1.96L_{Bp}^{1.36})$
Pandolf	≥ 0	$Watts = 1.5m + 2.0(m + l)(l/m)^2 + t(m + l)(1.5s^2 + 0.35sg)$
	< 0	$Watts = [1.5m + 2.0(m + l)(l/m)^2 + t(m + l)(1.5s^2 + 0.35sg)] - [t(g(m + l)s/3.5 - (m + l)(g + 6)^2/m + 25 - s^2)]$
Dill	≥ 0	$VO_2 = 3.5 + t(0.1\sigma + 1.8\sigma\gamma)$
	< 0	$VO_2 = 3.5 + 0.73t(0.1\sigma)$
de Müllenheim	≥ 0	$VO_2 = 4.0472 + t(2.0167v + 0.293vg)$
	< 0	$VO_2 = 4.0472 + 0.73t(2.0167v)$

Table 1. Predictive formulas expressed in original energy units. VO_2 = oxygen consumption (mL/kg/min), RMR = resting metabolic rate (mL/kg/min), M_{rest} = resting metabolic rate (Watts/kg), L_{Bp} = backpack load (kg)/body mass (kg), t = terrain coefficient, g = %grade (0-100), s = speed (m/s), m = body mass (kg), l = load (kg), σ = speed (m/min), γ = %grade (0-1), v = speed (km/h).

	Grades	Equation
Ludlow	≥ 0	$VO_2 = RMR + [(m + l)/m]t[0.32g + 3.28 + (1 + 0.19g)2.66s^2]$
	< 0	$VO_2 = RMR + 0.73[(m + l)/m]t[0.32g + 3.28 + 2.66s^2]$
Looney		$VO_2 = (60/20.1)/m \{ RMR/(60/20.1) + [0.19 + t(1.78s^{0.58} + 0.27s^4 + (0.34sg(1 - 1.05^{1 - 1.1^{(g+32)})))](1 + 1.96(l/m)^{1.36}) \}$
Pandolf	≥ 0	$VO_2 = (60/20.1)/m[1.5m + 2(m + l)(l/m)^2 + t(m + l)(1.5s^2 + 0.35sg)]$
	< 0	$VO_2 = (60/20.1)/m \{ [1.5m + 2(m + l)(l/m)^2 + t(m + l)(1.5s^2 + 0.35sg)] - [t(g(m + l)s/3.5 - (m + l)(g + 6)^2/m + 25 - s^2)] \}$
Dill	≥ 0	$VO_2 = 3.5 + t[0.1(60s) + 1.8(60s)(g/100)]$
	< 0	$VO_2 = 3.5 + 0.73t[0.1(60s)]$
de Müllenheim	≥ 0	$VO_2 = 4.0472 + t[2.0167(3.6s) + 0.2938(3.6s)g]$
	< 0	$VO_2 = 4.0472 + 0.73t[2.0167(3.6s)]$

Table 2. Predictive formulas expressed in VO_2 . VO_2 = oxygen consumption (mL/kg/min), RMR = resting metabolic rate (mL/kg/min), t = terrain coefficient, g = %grade (0-100), s = speed (m/s), m = body mass (kg), l = load (kg). A conversion from energy units to O_2 was done for the Looney and Pandolf equations using a conversion factor of 20.1 Joules per mL of O_2 (Blaxter K., 1989). This same conversion factor was used to convert from O_2 to energy units for resting metabolic rate in the Looney equation. Speed and gradient were converted from σ to s and γ to g respectively in the Dill equation to simplify the number of needed variables. L_{Bp} was also simplified to (l/m) for the Looney equation, and v was simplified to s for the de Müllenheim equation.

Age	Men	Women
18-30	$RMR = 2.177 + (100.055 / m)$	$RMR = 2.142 + (70.343 / m)$

Table 3. Schofield RMR equations for men and women ages 18-30 yrs as presented by Weyand et al. (2021). RMR = resting metabolic rate (mL/kg/min), m = body mass (kg).

Two equations directly incorporate load (Looney et al., 2022a; Pandolf et al., 1977), while Ludlow and Weyand (2017) proposed multiplying the active portion of their equation by a factor of the load added. Ludlow and Weyand argue that incorporating total load into the whole equation over inflates the value given from the resting portion of energy expenditure. The Dill and de Müllenheim equations do not incorporate load, so it will only be used in the other three equations (de Müllenheim et al., 2016; Dill, 1965).

The equation from Ludlow and Weyand (2017) requires a person's RMR (resting metabolic rate) in their model. Estimated RMR was utilized using the Schofield et al. (1985) equation (see Table 3) since most potential users would not have the knowledge or ability to measure this value directly. Weyand et al. (2013) suggested using the Schofield et al. (1985) RMR estimation in their original formulation of their model, and we used their version with the incorporated conversions from MJ/24h to VO_2 (Weyand et al., 2021).

Another factor that may affect the accuracy of equations is the difference in surface characteristics. It is widely accepted that multiplying the walking portion of the equations by a terrain coefficient is adequate enough to account for any differences that this may impart on accuracy (Looney et al., 2019; Pandolf et al., 1977; Richmond et al. 2015). However, it is important to note that while this works well for those equations that separate the moving portion of their model from the standing portion like in Looney et al. (2022a), the effect of the terrain coefficient might be over inflated in equations like the Ludlow model since their active portion includes standing energy expenditure, which terrain does not affect (Ludlow & Weyand, 2017). Nevertheless, the terrain coefficient was applied uniformly across all equations. For this study the terrain coefficient of 1.2 - provided by Richmond et al. (2015) for gravel/dirt - was utilized based on the trail characteristics.

It has also been shown that the use of walking/hiking sticks increases the overall energy expenditure of walking (Saunders et al., 2008), so their use was not allowed within this study.

The respiratory exchange ratio (RER) associated with energy usage during exercise changes depending on how long ago an individual has eaten a meal (Montain et al., 1991). Due to this, participants were asked to eat a meal 3-4 hours before arriving to participate in the study,

and were not allowed to eat during the hike, so that there wasn't any variance in energy usage due to meal times affecting RER.

Research Design Overview

The research design used was a prospective comparative validation study using a convenience sample of avid hikers.

Participants

Fifteen subjects (10 male, 5 female) were recruited based on the G*power analysis of a previous validation study on energy expenditure equations (Weyand et al., 2021) with an alpha level of 0.05, power of 0.95, 1 group and 6 measurements set, an effect size of 0.25, a correlation among repeated measures of 0.75 and a non-sphericity correction set to 1. It was expected that a larger effect size would have been warranted based on those found in Weyand et al. (2021), but a smaller effect size was used to avoid underestimating the number of needed participants. This was also true of the correlation among repeated measures; the equations predict the same thing, so they should have a higher correlation (e.g., $>.90$), but a lower value was used to guarantee enough participants were recruited. Inclusion criteria were: ages 18-42 y, based on the ages of participants from which the equations were developed, be avid hikers (i.e., hike 1-2 times a month), and able to pass the Get Active Questionnaire. Exclusion criteria were: no recent lower limb injuries, either within the last 6 months, or that they were still recovering from if longer; and no respiratory diseases or abnormalities, or the cold or flu within the last 2 weeks (Gumede et al., 2024; Ismail & Schellack, 2017). IRB approval was acquired before starting recruitment, and participants were recruited by word of mouth and posted flyers on a university campus. Participants reviewed the informed consent, answered eligibility questions, scheduled a day and

time for data collection, and received basic preliminary instruction online via email communication. Informed consent was obtained from participants in person at the trailhead.

Preliminary Procedures

GPS data from a Polar V800 watch (Polar Electro, Kempele, Finland), which has been validated as one of the most accurate GPS watches (Gilgen-Ammann et al., 2020), was used to measure a course profile of the Wind Caves trail, a popular hiking trail in northern Utah. The distance and elevation profile are provided in Figure 1, and grade, distance, and average speed data in Table 4. Trail profile was used to obtain the gradient data needed for the equations, both overall and for each of four trail sections. This was done by the researchers prior to any data collection. The Wind Caves trail was divided into four sections based on differing grades called: the flats, hill climber 101, switch backs, and the landing. Both average gradient data over the whole up- and down-hill portions were used since that is what would be most widely used by individuals in the field, and it was also obtained for each of the four sections since the variety of different gradients may have impacted equation accuracy.

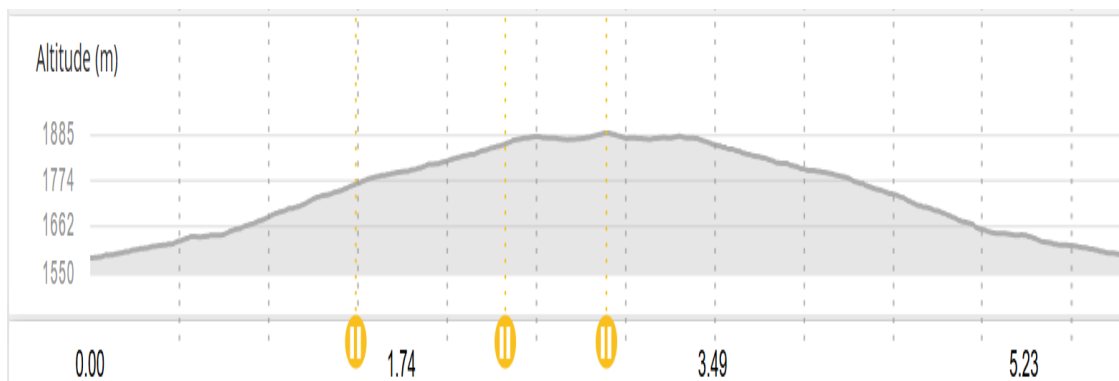


Figure 1. Gradient profile over the whole trail. Y-axis in meters. X-axis in kilometers.

	Gradient	Distance	Speed
Total Up	10.63%	2.88	1.37
1-2	9.13%	0.92	1.47
2-3	17.33%	0.60	1.17
3-4	11.50%	0.80	1.35
4-5	4.64%	0.56	1.52
Total Down	-10.85%	2.84	1.45
5-6	-4.36%	0.55	1.48
6-7	-11.63%	0.80	1.51
7-8	-17.97%	0.59	1.30
8-9	-9.44%	0.90	1.51

Table 4. Average gradient (%), distance (km), and speed (m/s) over the entire uphill trail, downhill trail, and for each trail section as measured by Polar V800 GPS watch.

In the case of rain/inclement weather, any testing was postponed both for safety considerations as well as the change in terrain coefficient that this imparted. Data collection was also postponed if temperatures fell outside of 15-32°C (59-90°F) for the safety of participants. Participants were notified by email using standardized communication at least 24 hours in advance if the weather forecast showed these scenarios during the scheduled data collection time, and then data collection was rescheduled.

Protocol

Calibration of the Cosmed K5 mobile metabolic unit (Cosmed USA, Inc., Concord, CA) occurred at the trailhead according to manufacturer recommendations using a 3-L syringe and calibration gas of 16% oxygen and 5% carbon dioxide (*K5 User Manual*, 2015). The K5 has been shown to be valid and reliable (DeBlois et al., 2021; Guidetti et al., 2018; Perez-Suarez et al., 2018), with the minimum detectable change ranging from 2-3.8% (Guidetti et al., 2018). This represents the magnitude of change needed to know it wasn't just due to measurement error.

Once the participant was at the trailhead, their informed consent was obtained and they were asked if anything had changed since their prescreening; their height and body mass were then measured and recorded once it had been determined that they still

qualified. Body mass was measured using a digital scale (model 869, Seca, Chino, CA) with participants wearing normal hiking clothes of the participant's choosing including shoes. Height was measured with participants laying down on a yoga mat using a tape measure. Pack weight was calculated as 5% of the participant's body mass. A Camelbak Classic 70 oz Hydration Pack (Camelbak Products, LLC, Petaluma, CA) with the hydration system removed was loaded with weights so that the combined weight of the pack and K5 was 5% of the participant's body mass, rounded to the nearest 0.25kg. This weight was chosen as it is an approximate weight of a typical daypack, and it is also the amount of weight that can be added without affecting postural stability – which should be avoided to prevent a fall/injury (Daffin et al., 2023). Current time and date were recorded so that weather conditions could be recorded from climate.usu.edu using the Logan Golf Station AGWX from the 13 AGWX (Utah AgWeather) network as the recording station.

The participant was fitted with the weighted pack along with the K5 unit. Once data collection was started in breath-by-breath mode simultaneously along with a stopwatch, the participant hiked along the designated trail at a self-selected walking pace. No running was allowed as it is not what the equations were designed to predict. No eating during the hike was allowed as that would have affected participant RER. Average speed for the whole trail and each section was calculated post-hike using time and distance data. A researcher trailed behind the participant at a moderate distance of 3-6 m (10-20 ft): far enough so as not to influence speed but close enough to assist with pausing data collection and handling equipment if needed. The researcher also carried other necessary supplies like water, a data collection sheet for times, a first-aid kit, and

a mini-inReach for themselves and the participant. If needed, the participant was allowed to stop for rest and/or water and was asked to turn around to clearly indicate to the researcher that assistance was required. If so, the researcher marked that time point with the stopwatch and helped remove the K5 mask so that the participant could drink. After any rests, the researcher helped reattach the mask and marked the time, and the participant then continued hiking. Along the course, the researcher marked the time as the participant passed the transition points between trail sections, which were marked and known to the researcher. Once the turnaround point had been reached, the researcher marked the time, and the participant continued back down the trail the same way they came up, with the researcher marking the time as they passed certain points along the trail again. Once the bottom was reached, the researcher marked the time, stopped data collection, and then helped the participant take off the K5 and pack. All participants hiked the same trail with no variability of grade or distance.

Complimentary snacks (granola bars, candy, etc.) were offered after data collection as a courtesy to participants, and they were instructed to use their own discretion as to whether or not they could eat the provided snacks.

Data Analysis

Data were downloaded from the K5 unit to the manufacturer's software program (Cosmed OMNIA 1.6.10) for data analysis after each day of data collection.

Once data for all participants were collected, the breath-by-breath data for the entire uphill trail, entire downhill trail, and for each individual trail section was averaged. Data that fell within a break where the K5 mask was removed was excluded. Researchers performed correlation, repeated measures ANOVA, and equivalence testing analyses between the average

measured values from the K5 and those calculated by the energy expenditure equations. Data analysis was done in Jamovi (ver. 2.3.28). A correlation between each equation and the measured values was run to see how much variance in the estimated values is explained by the variance in the measured values. Researchers also tested to see if there was a statistically significant difference between the measured and estimated values using ANOVA with a significance level of $p < .05$. Researchers also ran an equivalence test if no statistically significant difference was found with a smallest effect size of interest (SESOI) equal to plus or minus 10% (Looney et al., 2024) to see if the estimated values were equivalent to the measured values in a practical way.

These analyses were done separately for uphill and downhill components of the hike. The measured values were also compared to both the values calculated using average gradients over the entire up- and down-hill portions, as well as values calculated from the four different sections of the trail with varying gradient values.

Results

All of the prediction equations were extremely highly correlated with one another, ranging from a Pearson's r of >0.99 down to a Pearson's r of 0.94 ; all had a p -value $<.001$ (see tables A5-A14). When comparing the prediction equations to the measured values however, the correlation coefficients tended to be smaller, though they were still very highly correlated. For the uphill comparisons, they had a Pearson's r of up to 0.94 ($p<.001$) in table 8 between every equation and measured values and table 9 between the Ludlow, Looney and Pandolf equations and measured values, down to a Pearson's r of 0.88 ($p<.001$) in table 6 between the Looney equation and measured values. In the downhill comparisons, they had a Pearson's r of up to 0.86

($p < .001$) between the measured values and the Looney equation in Table 11, down to 0.64

($p = .010$) in table 13 between the Pandolf equation and measured values.

After running a repeated measures ANOVA (see table A15), post-hoc pairwise comparisons with Tukey adjustment showed that the predicted VO_2 values were not statistically significantly different from the reference measured VO_2 in quite a few cases ($p > 0.05$); see Table 16.

	mean difference	standard error	t(14)	p
Ludlow 1-2	-0.22	0.80	-0.27	>0.999
Looney Up	1.25	0.69	1.82	0.486
Looney 1-2	-1.68	0.85	-1.97	0.407
Looney 2-3	0.62	0.88	0.70	0.979
Looney 3-4	1.61	0.72	2.25	0.275
Pandolf Up	-0.08	0.72	-0.11	>0.999
Pandolf 2-3	0.26	0.94	0.28	>0.999
Pandolf 3-4	0.48	0.73	0.66	0.984
Pandolf 4-5	1.90	0.64	2.96	0.087
Pandolf 6-7	1.00	0.61	1.63	0.596
Pandolf 7-8	-0.45	0.64	-0.70	0.979
Pandolf 8-9	1.31	0.61	2.16	0.312
Dill 1-2	-0.18	0.66	-0.27	>0.999
Dill 2-3	1.27	0.92	1.38	0.738
de Müllenheim Up	0.04	0.65	0.05	>0.999
de Müllenheim 2-3	-0.50	0.91	-0.55	0.993
de Müllenheim 3-4	0.50	0.62	0.81	0.962
de Müllenheim 6-7	1.28	0.64	2.00	0.391
de Müllenheim 8-9	1.43	0.60	2.40	0.220

Table 16. Post-hoc pairwise comparisons with Tukey adjustment between the equations and measured values when there was no statistically significant difference.

The mean difference between the predicted and measured values are shown in Table 17. The Ludlow equation was not statistically significantly different only on the trail section between markers 1 and 2. The Looney equation was not statistically significantly different on every uphill portion except for the trail between the 4th and 5th markers. The Pandolf equation was not statistically significantly different on all uphill trail portions except for between the 1st and 2nd markers. The Pandolf equation was also not statistically significantly different on the last 3

downhill trail sections. The Dill equation was not statistically significantly different on the trail section between markers 1 and 2, and 2 and 3. The de Müllenheim equations was not statistically significantly different on the overall uphill trail section and on the trail sections between markers 2 and 3, and 3 and 4. The de Müllenheim equation was also not statistically significantly different on downhill trail sections between the 6th and 7th markers as well as the 8th and 9th markers.

Difference	Measured Values	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Total Up	34.52	-3.13 9.08%	-1.25 3.63%	0.08 0.23%	-2.27 6.56%	-0.03 0.10%
1-2	31.39	0.22 0.70%	1.68 5.34%	3.40 10.83%	0.18 0.57%	2.60 8.27%
2-3	39.52	-4.51 11.42%	-0.62 1.57%	-0.23 0.65%	-1.27 3.22%	0.50 1.28%
3-4	35.97	-3.80 10.55%	-1.61 4.48%	-0.48 1.34%	-2.67 7.43%	-0.50 1.39%
4-5	29.16	-4.73 16.22%	-3.35 11.48%	-1.90 6.50%	-5.55 19.02%	-2.89 9.90%
Total Down	16.38	-4.85 29.61%	-4.43 27.05%	-3.40 20.79%	-5.25 32.07%	-3.10 18.95%
5-6	21.43	-9.67 45.12%	-6.80 31.75%	-6.48 30.23%	-10.13 47.29%	-7.95 37.10%
6-7	14.93	-2.98 19.95%	-2.18 14.64%	-1.00 6.69%	-3.50 23.42%	-1.28 8.60%
7-8	14.36	-3.83 26.64%	-3.05 21.23%	0.45 3.12%	-4.01 27.93%	-2.03 14.10%
8-9	15.11	-3.12 20.62%	-2.28 15.12%	-1.31 8.69%	-3.65 24.14%	-1.43 9.45%

Table 17. Difference between average predicted VO₂ and average measured VO₂ (in mL/kg/min and %), along with average measured values (mL/kg/min) on the left.

Equivalence testing on the equations that weren't statistically significantly different showed that during the downhill portions of the trail, none of the equations were equivalent to measured VO₂, with equivalence bounds set a priori as $\pm 10\%$ (Looney et al., 2024) of the criterion mean difference. During the uphill trail portions, the Ludlow equation showed equivalence to measured values during the one section where it wasn't statistically significantly

different. The Looney equation showed equivalence during the overall downhill portion as well as during the trail section between the 2nd and 3rd, and 3rd and 4th markers, as did the Pandolf and de Müllenheim equations. The Dill equation showed equivalence to measured values on the trail section between markers 1 and 2 as well as the section between markers 2 and 3.

Discussion

All of the equations showed equivalence to the measured VO₂ values during at least one trail section, all occurring during the uphill portions of the trail. None of the equations showed equivalence to measured values on the downhill trail portions. Of the times they were equivalent, the Ludlow equation was only equivalent during a single section of trail, the Dill equation was equivalent over two different sections, and the Looney, Pandolf, and de Müllenheim equations were all equivalent to measured values during two trail sections as well as for the overall uphill.

Given this information, it would appear as though the Looney, Pandolf, and de Müllenheim equations are the best options for field use. Between those three, it could be speculated that the Pandolf equation is the most accurate, since it was not statistically significantly different from measured values over the most trail sections, followed by de Müllenheim (see Table 16). However, since these equations didn't show equivalence over more trail sections, this interpretation should be used with caution.

It wasn't surprising that the Pandolf and Looney equations did a little better than the Dill and Ludlow equations given that they were originally formulated to account for both load and grade. It was surprising how well the de Müllenheim equation did however, since it has no way to account for load.

One factor that may have influenced the predicted values was how participant characteristics were gathered. Participant mass was measured with hiking clothes on, including

shoes. This could have inflated the predicted values by a little bit, since actual participant mass was likely lower than what was recorded. Given that the prediction equations underestimated the VO_2 values in most cases, this likely didn't affect their accuracy in a meaningful way.

The participants were carrying a light load, and the study was also conducted at moderate altitude (1500-1800m), both of which could have increased participant workload. If either of these things weren't adequately accounted for by the equations, this could have increased the difference between the predicted and measured values.

One thing that could have made the Pandolf equation better at estimating downhill VO_2 was the Santee correction factor, and how it approached downhill grades in a more nuanced way. It treated steeper downhill grades differently than shallower ones, using the "J-shaped" curve (Looney et al., 2019; Santee et al., 2003). This doesn't account for the fact that the Looney equation also incorporated downhill grades in a similar fashion yet also did poorly in predicting downhill VO_2 . This interpretation should also be used cautiously, since the Pandolf equation only showed that it wasn't different from measured values on some of the downhill portions, not that it was equivalent.

Some limitations of this study are that it was conducted at a moderate altitude, which could have increased participant workload as mentioned above. Another limitation is the population of young adults. Both of these could potentially limit the generalizability of these findings to other settings/populations. Having this study out in the field is also one of its strengths, as this increases the generalizability of the study to more realistic hiking scenarios more so than previously done lab-based studies. Another limitation of our study design that should be accounted for in future research was not having a rest period for participants in between the uphill and downhill portions of the hike. Most of the participants got to the top of

the trail and immediately turned around to go downhill. This likely caused a carryover effect, making our measured VO_2 values during the first section of the downhill trail between markers 5 and 6 more elevated than they should have been, thus creating a bigger difference between it and the predictions. This can be seen in Table 16, likely explaining the larger difference between measured and predicted VO_2 across all equations.

The prediction equations' inability to accurately predict downhill VO_2 shows that their predictions should be used with caution in field-based settings, where most scenarios would require both uphill and downhill travel. The Pandolf, de Müllenheim, and Looney equations showed equivalence over more trail sections than the Dill and Ludlow equations, at least during the uphill trail portions, making them the best options to choose. The Pandolf equation also showed fewer statistically significant differences than the rest and is therefore the best candidate for field use with grade and load being taken into account.

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Appendix

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.93***	-				
Looney	0.93***	>0.99***	-			
Pandolf	0.93***	>0.99***	>0.99***	-		
Dill	0.93***	>0.99***	0.99***	>0.99***	-	
de Müllenheim	0.93***	>0.99***	0.99***	>0.99***	>0.99***	-

*** p < .001

Table 5. Pearson's r correlation matrix for the entire uphill trail.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.89***	-				
Looney	0.88***	>0.99***	-			
Pandolf	0.89***	>0.99***	>0.99***	-		
Dill	0.89***	>0.99***	0.99***	>0.99***	-	
de Müllenheim	0.89***	>0.99***	0.99***	>0.99***	>0.99***	-

*** p < .001

Table 6. Pearson's r correlation matrix for trail section 1-2.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.92***	-				
Looney	0.92***	>0.99***	-			
Pandolf	0.91***	>0.99***	>0.99***	-		
Dill	0.91***	0.99***	>0.99***	>0.99***	-	
de Müllenheim	0.91***	0.99***	>0.99***	>0.99***	>0.99***	-

*** p < .001

Table 7. Pearson's r correlation matrix for trail section 2-3.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.94***	-				
Looney	0.94***	>0.99***	-			
Pandolf	0.94***	>0.99***	>0.99***	-		
Dill	0.94***	>0.99***	0.99***	>0.99***	-	
de Müllenheim	0.94***	>0.99***	0.99***	>0.99***	>0.99***	-

*** p < .001

Table 8. Pearson's r correlation matrix for trail section 3-4.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.94***	-				
Looney	0.94***	>0.99***	-			
Pandolf	0.94***	>0.99***	>0.99***	-		
Dill	0.93***	>0.99***	0.99***	>0.99***	-	
de Müllenheim	0.93***	>0.99***	0.99***	>0.99***	>0.99***	-

*** p < .001

Table 9. Pearson's r correlation matrix for trail section 4-5.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.75 p = .001	-				
Looney	0.77***	0.99***	-			
Pandolf	0.76***	0.99***	0.99***	-		
Dill	0.74 p = .001	0.99***	0.98***	>0.99***	-	
de Müllenheim	0.74 p = .001	0.99***	0.98***	>0.99***	>0.99***	-

*** p < .001

Table 10. Pearson's r with p-values correlation matrix for the entire downhill trail.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.80***	-				
Looney	0.86***	0.99***	-			
Pandolf	0.80***	0.99***	0.98***	-		
Dill	0.75	0.99***	0.95***	0.99***	-	
	p = .001					
de Müllenheim	0.75	0.99***	0.95***	0.99***	>0.99***	-
	p = .001					

*** p < .001

Table 11. Pearson's r with p-values correlation matrix for trail section 5-6.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.77***	-				
Looney	0.82***	0.99***	-			
Pandolf	0.79***	>0.99***	0.99***	-		
Dill	0.76	0.99***	0.98***	>0.99***	-	
	p = .001					
de Müllenheim	0.76	0.99***	0.98***	>0.99***	>0.99***	-
	p = .001					

*** p < .001

Table 12. Pearson's r with p-values correlation matrix for trail section 6-7.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.70	-				
	p = .004					
Looney	0.73	0.99***	-			
	p = .002					
Pandolf	0.64	0.96***	0.95***	-		
	p = .010					
Dill	0.71	0.99***	0.98***	0.94***	-	
	p = .003					
de Müllenheim	0.71	0.99***	0.98***	0.94***	>0.99***	-
	p = .003					

*** p < .001

Table 13. Pearson's r with p-values correlation matrix for trail section 7-8.

Correlation Matrix	Measured	Ludlow	Looney	Pandolf	Dill	de Müllenheim
Measured	-					
Ludlow	0.78***	-				
Looney	0.81***	0.99***	-			
Pandolf	0.80***	0.99***	0.99***	-		
Dill	0.77***	>0.99***	0.98***	>0.99***	-	
de Müllenheim	0.77***	>0.99***	0.98***	>0.99***	>0.99***	-

*** p < .001

Table 14. Pearson's r with p-values correlation matrix for trail section 8-9.

ANOVA	F _(5,70)	p	η^2
Total Up	13.40	<.001	0.04
1-2	12.41	<.001	0.05
2-3	20.06	<.001	0.04
3-4	14.50	<.001	0.04
4-5	19.23	<.001	0.10
Total Down	21.74	<.001	0.26
5-6	40.93	<.001	0.40
6-7	11.04	<.001	0.14
7-8	38.09	<.001	0.36
8-9	11.40	<.001	0.15

Table 15. ANOVA results for the entire uphill trail, each uphill trail section, the entire downhill trail, and each downhill trail section.